Question			Answer	Marks	AO	Guidance	
10	(a)		gradient AC $= \frac{4-(-1)}{1-(-4)} = 1$	M1	2.1	Gradient calculation must be seen	
			A on the line so the equation of the line is $y-4=1(x-1)$			o.e. using C (–4, –1)	Also allow using $y = x + c$ and evaluating c
			so the equation is $y = x + 3$	A1	2.1	AG Must be clearly shown	
			Alternative method				
			When $x=1$ $y=1+3=4$ so A lies on the line				
			When $x=-4$ $y=-4+3=-1$ so C lies on the line	M1		checking both points lie on the line	
			So the equation of the line AC is $y = x + 3$	A1		clear conclusion must be seen	
				[2]			
10	(b)		M is (4, 2)	B1	1.1b		
			$y = x + 3$ crosses the x-axis at D (–3, 0)	B1	1.1b		
			length DM is $\sqrt{49+4} = \sqrt{53}$	M1	3.1a	Attempt to find at least two of the sides of triangle DMA	Also allow for the AC^2 etc found if clear
			length MA is $\sqrt{9+4} = \sqrt{13}$				
			length DA is $\sqrt{16+16} = \sqrt{32}$	A1	1.1b	At least 2 correct lengths soi FT their coordinates	
			using the cosine rule				Allow M1A0A0 for cosine rule used leading to one of the other two angles of the triangle. (101.3° or 29.1°) and not used to find 49.6°
			$\cos \hat{DMA} = \frac{13+53-32}{2\times\sqrt{13}\times\sqrt{53}}$	M1	1.1a	Allow sign errors Any form	
				A1	1.1b	FT their lengths. Must be correct expression for $\cos \hat{DMA}$	
			$\hat{DMA} = 49.6°$	A1	1.1b	Accept 49.7°	
				[7]			